

GEOS-MPM Verification Suite Report

Generated by verification/runSuite

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1 Running and rerunning

Cases are staged under `testRunDirectory/verification/<case>`. The suite records failures and continues unless `--stop-on-error` is supplied. Use `--force` for noninteractive reruns.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter
verification/runSuite run --python /usr/tce/bin/python3 --force --jobs 4 --case-jobs 2
verification/runSuite report --python /usr/tce/bin/python3
```

2 Timing and dispatch summary

Case	Status	Family	GEOS job(s)	GEOS state	GEOS wall	Issue
surfaceInformedPolymerLayer	failed	surfaceInformedPolymerLayer	6281619, 6281620, 6281621, 6281622	COMPLETED	Continuum tension: 00:00:03; compression: 00:00:03; Cohesive-zone tension: 00:00:03; Cohesive-zone compression: 00:09:52	timeout

Subcase timing detail. Dispatch time is local wall time spent staging inputs, running PFW, and submitting the GEOS job. GEOS wall and CPU columns are populated from **sacct** when available.

Case	Subcase	GEOS job	State	GEOS wall	Total CPU	NCPUS	Dispatch
surfaceInformedPolymerLayer	Continuum tension	6281619	COMPLETED	00:00:03	00:00:00	1	2.59
surfaceInformedPolymerLayer	Continuum compression	6281620	COMPLETED	00:00:03	00:00:00	1	2.532
surfaceInformedPolymerLayer	Cohesive-zone tension	6281621	COMPLETED	00:00:03	00:00:00	1	2.326
surfaceInformedPolymerLayer	Cohesive-zone compression	6281622	COMPLETED	00:09:52	00:00:00	1	2.175

3 Failures, expected failures, and diagnostics

3.1 surfaceInformedPolymerLayer

Status: failed Issue: timeout

```
--- surfaceInformedPolymerLayer_cohesive_tension_visit_render.log ---
Traceback (most recent call last):
--- surfaceInformedPolymerLayer_continuum_compression_visit_render.log ---
Traceback (most recent call last):
--- surfaceInformedPolymerLayer_continuum_tension_visit_render.log ---
Traceback (most recent call last):
```

4 Case studies

4.1 surfaceInformedPolymerLayer

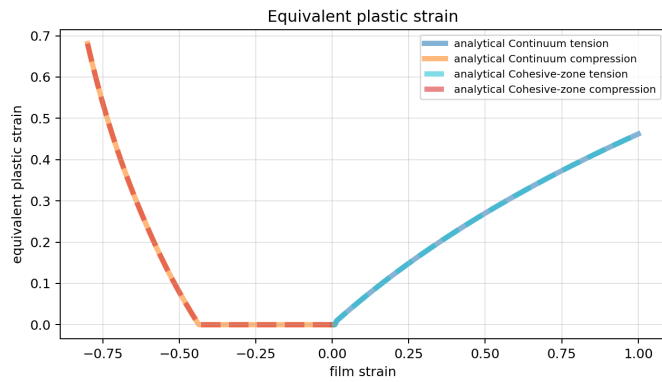
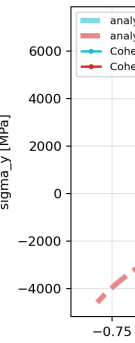
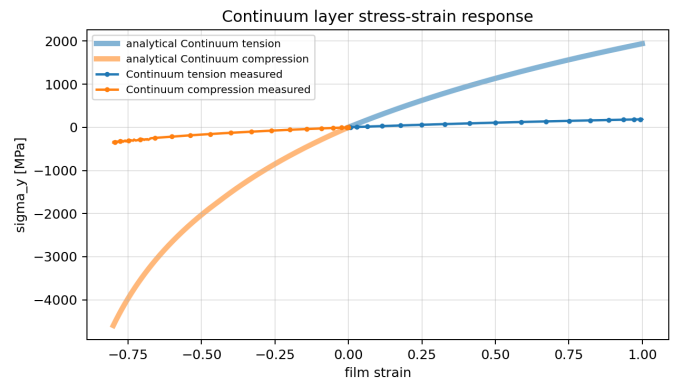
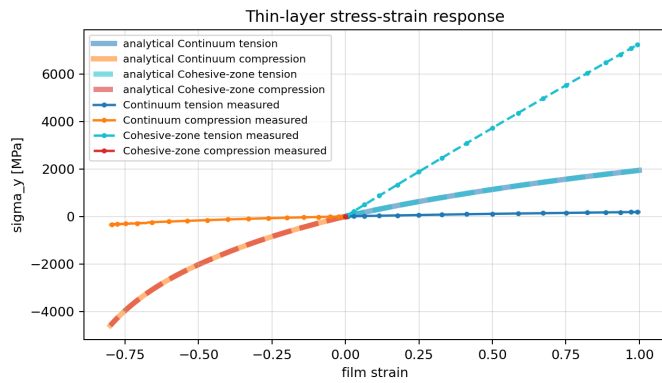
4.2 Surface-informed polymer layer

This verification folder exercises the surface-informed polymer as both a continuum material and a finite-thickness cohesive-zone material. Four variants are run from one input: continuum tension, continuum compression, cohesive-zone tension, and cohesive-zone compression. The continuum variants place a continuum polymer layer between stiff elastic loading bars and tie the two bar/polymer interfaces with stiff linear elastic cohesive zones. Those tie zones are not polymer material models; they are numerical bonded-interface fixtures that transmit load without relying on mixed continuum cells at the bar/polymer material interfaces. The cohesive-zone variants retain two stiff elastic blocks separated by a finite-thickness polymer cohesive interface with the same film thickness. The imposed cosine ramp is intentionally slow compared with the layer acoustic-transit time so the history is a quasi-static constitutive check rather than a wave-propagation benchmark.

The comparison curves use the continuum J2 radial-return update with retained mean stress for continuum variants and the reduced cohesive thin-film update with the same retained volumetric normal stress for cohesive-zone variants. In the cohesive projection, the mean film stress $p = K\epsilon_n$ is retained and only the deviatoric normal and shear stresses are returned to the scalar flow surface. The verification uses reusable non-polymer `ElasticIsotropic` block and `UncoupledCohesiveZone` tie cards from `pfw_materials.py`; the polymer response is carried by the continuum polymer layer in the continuum variants and by the polymer cohesive law in the cohesive-zone variants. The input explicitly requests `particleStress`, `particleDamage`, and `particlePlasticStrain` through `plottableFields`. The equivalent-plastic-strain visualization is a VisIt/post-processing expression computed from the primitive plastic-strain tensor components rather than a derived solver field. For measured histories, the film-strain coordinate is reconstructed from the actual deformation-gradient or length history written by GEOS when available, with the input cosine loading table used only as a fallback. When VisIt is available, it also renders loading-direction stress and equivalent plastic strain at four output states for each variant.

Quantitative result. The continuum variants place a finite-thickness continuum polymer layer between elastic loading bars; the cohesive-zone variants place a finite-thickness polymer CZ between elastic blocks. Both measured curves use boundary reactions. The analytical comparison uses the attached surface-polymer curve driver in confined thin-film mode at 21°C and 7.4% crystallinity; the driver integrates true strain and the plots report engineering film strain for comparison with the

	Variant	samples	RMS stress error	max stress error	final stress
GEOS deformation history. Stresses are reported in MPa.	Continuum tension	3e+02	1133.45	1754.48	186.45
	Continuum compression	3e+02	2249.98	4258.58	-353.30
	Cohesive-zone tension	3e+02	3233.72	5335.12	7275.9
	Cohesive-zone compression	3	0.280227	0.48532	-0.57889



```

--- surfaceInformedPolymerLayer_cohesive_tension_visit_render.log ---
Traceback (most recent call last):
--- surfaceInformedPolymerLayer_continuum_compression_visit_render.log ---
Traceback (most recent call last):
--- surfaceInformedPolymerLayer_continuum_tension_visit_render.log ---
Traceback (most recent call last):

```